

Red Hat Lightspeed 1-latest

Converting from a Linux distribution to RHEL using the Convert2RHEL utility in Red Hat Lightspeed

Instructions for a conversion from CentOS Linux 7 to Red Hat Enterprise Linux 7 using the Convert2RHEL utility in Red Hat Lightspeed

Last Updated: 2026-09-24

Red Hat Lightspeed 1-latest Converting from a Linux distribution to RHEL using the Convert2RHEL utility in Red Hat Lightspeed

Instructions for a conversion from CentOS Linux 7 to Red Hat Enterprise Linux 7 using the Convert2RHEL utility in Red Hat Lightspeed

Legal Notice

Copyright © Red Hat.

Except as otherwise noted below, the text of and illustrations in this documentation are licensed by Red Hat under the Creative Commons Attribution–Share Alike 3.0 Unported license . If you distribute this document or an adaptation of it, you must provide the URL for the original version.

Red Hat, as the licensor of this document, waives the right to enforce, and agrees not to assert, Section 4d of CC-BY-SA to the fullest extent permitted by applicable law.

Red Hat, the Red Hat logo, JBoss, Hibernate, and RHCE are trademarks or registered trademarks of Red Hat, LLC. or its subsidiaries in the United States and other countries.

Linux ® is the registered trademark of Linus Torvalds in the United States and other countries.

XFS is a trademark or registered trademark of Hewlett Packard Enterprise Development LP or its subsidiaries in the United States and other countries.

The OpenStack ® Word Mark and OpenStack logo are trademarks or registered trademarks of the Linux Foundation, used under license.

All other trademarks are the property of their respective owners.

Abstract

This document provides instructions on how to convert your operating system from CentOS Linux to RHEL 7 using the Convert2RHEL utility in Red Hat Lightspeed.

Table of Contents

PREFACE ... 3

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION ... 4

KEY MIGRATION TERMINOLOGY ... 5

CHAPTER 1. SUPPORTED CONVERSION PATHS ... 6

CHAPTER 2. CONVERSION METHODS ... 8

CHAPTER 3. PLAN A RHEL CONVERSION ... 9

CHAPTER 4. PREPARATION FOR A RHEL CONVERSION BY USING RED HAT LIGHTSPEED ... 11
4.1. PREREQUISITES 11
4.2. PREPARE FOR A RHEL CONVERSION USING RED HAT LIGHTSPEED 11
4.3. PREPARE FOR A RHEL CONVERSION WITH A PROXY SERVER 13
4.4. ADDITIONAL RESOURCES 14

CHAPTER 5. REVIEW THE PRE-CONVERSION ANALYSIS REPORT USING RED HAT LIGHTSPEED ... 16

CHAPTER 6. CONVERT TO A RHEL SYSTEM BY USING RED HAT LIGHTSPEED ... 18

CHAPTER 7. TROUBLESHOOTING CONVERSIONS ... 20
7.1. TROUBLESHOOTING RESOURCES 20
7.2. FIX DEPENDENCY ERRORS 20
7.3. TROUBLESHOOTING ISSUES WITH RED HAT LIGHTSPEED CONVERSIONS 20
7.3.1. Fix missing systems in pre-conversion analysis task 20
7.3.2. Pre-conversion analysis task fails to complete 21
7.4. KNOWN ISSUES AND LIMITATIONS 22
7.5. OBTAINING SUPPORT 24

CHAPTER 8. ADDITIONAL RESOURCES ... 25

PREFACE

This document provides instructions on how to convert your operating system from CentOS Linux to Red Hat Enterprise Linux (RHEL) 7 using Red Hat Lightspeed.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We are committed to providing high-quality documentation and value your feedback. To help us improve, you can submit suggestions or report errors through the Red Hat Jira tracking system.

Procedure

1. Log in to the Jira website.
   If you do not have an account, select the option to create one.
2. Click **Create** in the top navigation bar.
3. Enter a descriptive title in the **Summary** field.
4. Enter your suggestion for improvement in the **Description** field. Include links to the relevant parts of the documentation.
5. Click **Create** at the bottom of the dialogue.

KEY MIGRATION TERMINOLOGY

While the following migration terms are commonly used in the software industry, these definitions are specific to Red Hat Enterprise Linux (RHEL).

Update

Sometimes called a software patch, an update is an addition to the current version of the application, operating system, or software that you are running. A software update addresses any issues or bugs to provide a better experience of working with the technology. In RHEL, an update relates to a minor release, for example, updating from RHEL 8.1 to 8.2.

Upgrade

An upgrade is when you replace the application, operating system, or software that you are currently running with a newer version. Typically, you first back up your data according to instructions from Red Hat. When you upgrade RHEL, you have two options:

- **In-place upgrade:** During an in-place upgrade, you replace the earlier version with the new version without removing the earlier version first. The installed applications and utilities, along with the configurations and preferences, are incorporated into the new version.
- **Clean install:** A clean install removes all traces of the previously installed operating system, system data, configurations, and applications and installs the latest version of the operating system. A clean install is ideal if you do not need any of the previous data or applications on your systems or if you are developing a new project that does not rely on prior builds.

Operating system conversion

A conversion is when you convert your operating system from a different Linux distribution to Red Hat Enterprise Linux. Typically, you first back up your data according to instructions from Red Hat.

Migration

Typically, a migration indicates a change of platform: software or hardware. Moving from Windows to Linux is a migration. Moving a user from one laptop to another or a company from one server to another is a migration. However, most migrations also involve upgrades, and sometimes the terms are used interchangeably.

- **Migration to RHEL:** Conversion of an existing operating system to RHEL
- **Migration across RHEL:** Upgrade from one version of RHEL to another

CHAPTER 1. SUPPORTED CONVERSION PATHS

Currently, it is possible to convert your systems from the following Linux distributions and versions to the corresponding minor version of RHEL listed in Table 1.1.



IMPORTANT

Red Hat recommends that you seek the support of **Red Hat Consulting services** to ensure that the conversion process is smooth.

Table 1.1. Supported conversion paths

Source OS	Source version	Target OS and version	Product Variant	Available Conversion Methods
Alma Linux	9.7	RHEL 9.7	N/A	Command line, Satellite
	8.10	RHEL 8.10	N/A	Command line, Satellite
CentOS Linux	8.5	RHEL 8.5	N/A	Command line, Satellite
	7.9	RHEL 7.9	Server	Command line, Satellite, Red Hat Lightspeed
Oracle Linux	9.7	RHEL 9.7	N/A	Command line, Satellite
	8.10	RHEL 8.10	N/A	Command line, Satellite
	7.9	RHEL 7.9	Server	Command line, Satellite
Rocky Linux	9.7	RHEL 9.7	N/A	Command line, Satellite
	8.10	RHEL 8.10	N/A	Command line, Satellite

Because the last available minor version of CentOS Linux is CentOS Linux 8.5, it is not possible to convert from CentOS Linux 8 directly to the latest available minor version of RHEL 8. Update your system to the latest version of RHEL after the conversion.

RHEL 7 reached the end of the Maintenance Support Phase on June 30, 2024. If you are converting to RHEL 7 and plan to stay on RHEL 7, buy the Extended Life Cycle Support (ELS) add-on subscription. If you plan to convert to RHEL 7 and then immediately upgrade to RHEL 8 or later, an ELS subscription is

not needed. Note that without ELS, you have limited support for RHEL 7, including for the upgrade from RHEL 7 to RHEL 8. For more information, see the **Red Hat Enterprise Linux Life Cycle** and the **Convert2RHEL Support Policy**.

In addition to the supported conversion paths, you can perform an unsupported conversion from Amazon Linux 2, CentOS Stream 8, CentOS Stream 9, and Scientific Linux 7 to RHEL. For information about unsupported conversions, see **How to perform an unsupported conversion from a RHEL-derived Linux distribution to RHEL**.

You can also perform an unsupported simultaneous upgrade and conversion from CentOS Stream 9 or Alma Linux 9 to RHEL 10. For more information about an unsupported upgrade and conversion, see **Performing an unsupported upgrade and conversion from a RHEL-derived Linux distribution to RHEL 10**.

For information about Red Hat's support policy for Linux distribution conversions, see **Convert2RHEL Support Policy**.

Additional resources

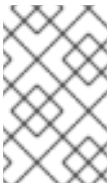
- Red Hat Consulting services
- Red Hat Enterprise Linux Life Cycle
- Convert2RHEL Support Policy
- How to perform an unsupported conversion from a RHEL-derived Linux distribution to RHEL
- Performing an unsupported upgrade and conversion from a RHEL-derived Linux distribution to RHEL 10

CHAPTER 2. CONVERSION METHODS

Depending on your requirements, you can convert your operating system to RHEL by using one of the various methods.

Converting by using the command line

Use this method to convert a small number of servers, or to convert a large number of servers at scale by using custom automation.



NOTE

If you have a Satellite subscription but plan to convert by using the command line, you can access the required RHEL packages through Satellite instead of using the Red Hat Content Delivery Network (CDN).

For more information about how to convert by using the command line, see **Planning a RHEL conversion** and **Converting using the command-line**.

Converting by using Red Hat Satellite

Use this method to convert a large number of servers at scale. To convert by using Satellite, you must have a Satellite subscription. For more information about how to convert by using Satellite, see *Converting a Host to Red Hat Enterprise Linux.

Converting by using Red Hat Lightspeed

Use this method to convert multiple systems in a user-friendly GUI interface without needing a Satellite subscription. Systems must be connected to the internet. Systems are fully onboarded with Red Hat Lightspeed after the conversion.



NOTE

Red Hat Lightspeed supports conversions with Red Hat Enterprise Linux for Third Party Migration with the pay-as-you-go pricing model on Amazon Web Services (AWS). For more information, see **Red Hat Enterprise Linux for Third Party Linux Migration**.

For more information about how to convert by using Red Hat Lightspeed, see **Planning a RHEL conversion** and **Converting using Red Hat Lightspeed**.

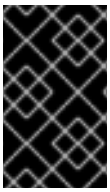
Additional resources

- Planning a RHEL conversion
- Converting using the command-line
- Converting using Red Hat Lightspeed
- Converting a Host to Red Hat Enterprise Linux
- Red Hat Enterprise Linux for Third Party Linux Migration

CHAPTER 3. PLAN A RHEL CONVERSION

You can perform the automated conversion process on a running system. The **Convert2RHEL** utility replaces all RPM packages from the original Linux distribution by their RHEL version. At the end of the process, you must restart the system to boot the RHEL kernel.

The conversion does not affect packages that are exclusive to the original distribution and do not have corresponding counterparts in RHEL repositories. It also does not affect the third-party packages, which originate neither from the original Linux distribution nor from RHEL. Red Hat does not provide support for third-party packages that are left unchanged during the conversion process. See the **Red Hat policy on supporting third-party software**.



IMPORTANT

The **Convert2RHEL** utility does not directly affect local users and their data in the **/home** and **/srv** directories. However, **Convert2RHEL** cannot control actions that RPM package scriptlets perform during the conversion process.

Consider the following before converting your system to RHEL:

Architecture

The source operating system must be installed on a system with 64-bit Intel architecture. It is not possible to convert with other system architectures.

Security

Systems in FIPS mode are not supported for conversion.

Kernel

Systems that are using kernel modules that do not exist in RHEL kernel modules are not currently supported for conversion. Red Hat recommends disabling or uninstalling foreign kernel modules before the conversion and then enabling or reinstalling those kernel modules afterward. Unsupported kernel modules include:

- Kernel modules for specialized applications, GPUs, network drivers, or storage drivers
- Custom compiled kernel modules built by DKMS

Public clouds

Conversions on public clouds are supported in the following situations:

- Alma Linux, CentOS Linux, and Rocky Linux - Use Red Hat Subscription Manager (RHSM) for the following:
  - Images on Amazon Web Services (AWS), Microsoft Azure, and Google Cloud with no associated software cost.
  - User-provided custom images on all public clouds
- Oracle Linux - Uses RHSM for user-provided custom images on all public clouds.

Convert2RHEL is unable to access RHEL packages through Red Hat Update Infrastructure (RHUI) during the conversion.

High Availability

Systems that are using high availability cluster software by Red Hat or third parties are not currently tested or supported for conversion to RHEL. Red Hat recommends migrating to newly installed RHEL systems to ensure the integrity of these environments.

Identity Management

Performing an in-place conversion of a FreeIPA server is not supported. For more information about how to migrate a FreeIPA deployment to IdM, see **Migrating to IdM on RHEL 7 from FreeIPA on non-RHEL Linux distributions**, Migrating to IdM on RHEL 8 from FreeIPA on non-RHEL Linux distributions*, and **Migrating to IdM on RHEL 9 from FreeIPA on non-RHEL Linux distributions**

Foreman

Conversions of systems that use Foreman with the Katello plugin are not supported. To perform a supported conversion, migrate to Red Hat Satellite first and then proceed with the conversion.

RAID

It is not possible to convert UEFI-based systems with **mdadm**-managed RAID devices.

Additional resources

- Red Hat policy on supporting third-party software
- Migrating to IdM on RHEL 7 from FreeIPA on non-RHEL Linux distributions
- Migrating to IdM on RHEL 8 from FreeIPA on non-RHEL Linux distributions
- Migrating to IdM on RHEL 9 from FreeIPA on non-RHEL Linux distributions

CHAPTER 4. PREPARATION FOR A RHEL CONVERSION BY USING RED HAT LIGHTSPEED

Before running the pre-conversion analysis by using Red Hat Lightspeed and performing the conversion, you must first complete all necessary preparation steps.

4.1. PREREQUISITES

- You have a RHEL subscription. You can obtain a subscription using one of the following methods:
  - Get a no-cost RHEL Developer Subscription for Individuals. The Developer Subscription is limited to sixteen servers.
  - Start a 60-day free trial for a RHEL subscription. You can cancel this trial at any time.
  - Contact the Red Hat Sales team for assistance. If you plan to convert a large number of servers, Red Hat recommends coordinating with the Sales team.

4.2. PREPARE FOR A RHEL CONVERSION USING RED HAT LIGHTSPEED

Prepare your system for the pre-conversion analysis by using Red Hat Lightspeed and then performing the conversion to RHEL.

Prerequisites

- You are planning to convert CentOS Linux 7 systems to Red Hat Enterprise Linux (RHEL) 7. You must convert from CentOS Linux 8 and other Linux distributions in the command-line or Satellite.
- You have an account and activation key at Red Hat Customer Portal . For more information, see Getting started with activation keys on the Hybrid Cloud Console .
- You have stopped important applications, database services, and any other services that store data to reduce the risk of data integrity issues.
- You have temporarily disabled antivirus software to prevent the conversion from failing.
- You have disabled or adequately reconfigured any configuration management system, such as Salt, Chef, Puppet, Ansible, to prevent restoring the original system.
- The **sos** package is installed. You must use this package to generate an **sosreport** that the opening a support case for the Red Hat Support team requires.
- You have enabled Simple Content Access (SCA). Red Hat accounts created after July 15, 2022 have SCA enabled by default.

Procedure

1. Back up your systems and verify that you can restore them if needed.
2. Check Known issues and limitations and verify that your system is supported for conversion. Apply workarounds where applicable.

3. Ensure that the standard CentOS Linux kernel is the booted kernel. If the kernel your system is booted into is not the standard kernel, change the default kernel to the standard kernel and reboot your system to apply the changes. For more information, see Making Persistent Changes to a GRUB Menu Using the grubby Tool.

4. If you are converting with a firewall or through a proxy server, ensure that you have access to the following connections:

   - https://cdn.redhat.com
   - https://cdn-public.redhat.com
   - https://subscription.rhsm.redhat.com - required only for systems with firewalls
   - https://*.akamaiedge.net - required only for systems with firewalls
   - https://cert.console.redhat.com

5. Install the Red Hat client tools:

   a. Download the Red Hat GPG key:

   ```
   # curl -o /etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release
   https://security.access.redhat.com/data/fd431d51.txt
   ```

   b. Install the **client-tools** repository file:

   ```
   # curl -o /etc/yum.repos.d/client-tools.repo https://cdn-
   public.redhat.com/content/public/repofiles/client-tools-for-rhel-7-server.repo
   ```

   c. Install the **client tools** packages:

   ```
   # yum -y install subscription-manager subscription-manager-rhsm-certificates rhc rhc-
   worker-script insights-client
   ```

6. Enable Remote Host Configuration, register your systems with Red Hat Subscription Manager (RHSM), and connect your systems with Red Hat Lightspeed:

   ```
   # rhc connect --activation-key <activation_key> --organization <organization_ID>
   ```

   Replace *organization_id* and *activation_key* with the organization ID and activation key from the Red Hat Customer Portal. For more information about Remote Host Configuration, see the Remote Host Configuration (rhc) Knowledgebase article.

Verification

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Systems**.
2. Verify that your CentOS Linux systems appear as expected.



NOTE

The only Red Hat Lightspeed service you can use with registered CentOS Linux systems is the RHEL conversion. All other Red Hat Lightspeed services are available only after the conversion to RHEL.

4.3. PREPARE FOR A RHEL CONVERSION WITH A PROXY SERVER

Before running the pre-conversion analysis for your operating system with a proxy server by using Red Hat Lightspeed to perform the conversion. To do so, you must configure the Remote Host Configuration (RHC) daemon and Red Hat Lightspeed to use this proxy server.

Prerequisites

- The **yum** command is configured to use HTTP proxy. For more information, see the Red Hat Knowledgebase solution How to enable Proxy Settings for Yum Command on RHEL? .
- The following packages are installed:
  - **subscription-manager**
  - **subscription-manager-rhsm-certificates**
  - **insights-client**
  - **rhc**
  - **rhc-worker-script**.

Procedure

1. Set the **http_proxy** environment variable in the command line:

   ```
   # export http_proxy='http://<proxy_hostname>:<proxy_port>'
   ```

   Where http://*<proxy_hostname>:<proxy_port>* is your proxy server.

2. Set the repository for the client tools:

   ```
   # curl -o /etc/yum.repos.d/client-tools-for-rhel-7-server.repo
   https://ftp.redhat.com/redhat/client-tools/client-tools-for-rhel-7-server.repo
   ```

3. Verify the downloaded client tools packages with the Red Hat GPG public key:

   ```
   # curl -o /etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release
   https://security.access.redhat.com/data/fd431d51.txt
   ```

4. Edit the **/etc/systemd/system/rhcd.service.d/proxy.conf** file for the RHC daemon to use your proxy server:

   ```
   [Service]
   Environment="HTTP_PROXY=http://<proxy_hostname>:<proxy_port>"
   Environment="HTTPS_PROXY=http://<proxy_hostname>:<proxy_port>"
   ```

   Where http://*<proxy hostname>:<proxy port>* is your proxy server.

5. Reload the RHC daemon to apply new configuration:

   ```
   # systemctl daemon-reload
   ```

6. Edit the **/etc/insights-client/insights-client.conf** file for Red Hat Lightspeed to use your proxy server:

   ```
   [insights-client]
   proxy=http://<proxy_hostname>:<proxy_port>
   ```

   Where http://*<proxy_hostname>:<proxy_port>* is your proxy server.

7. Edit the **/etc/rhsm/rhsm.conf** file to configure the Red Hat Subscription Manager to use a proxy server:

   ```
   proxy_hostname = <proxy_hostname>
   proxy_port = <proxy_port>
   proxy_scheme = http
   ```

   Where *<proxy_hostname>* and *<proxy_port>* are parameters of you proxy server.

8. Register your system with the Red Hat Subscription Manager and Red Hat Lightspeed:

   ```
   # rhc connect --activation-key <activation_key> --organization <organization_ID>
   ```

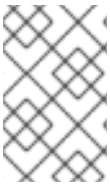
   Where the *<activation_key>* and the *<organization_ID>* are the activation key and organization ID from the Red Hat Customer Portal . For more information about Remote Host Configuration, see the link: Remote Host Configuration (rhc) Knowledgebase article.

Verification

1. Access the Red Hat Hybrid Cloud Console, and navigate to **Inventory → Systems**.
2. Verify that your CentOS Linux systems appear as expected.

Next steps

- Proceed to review the pre-conversion analysis report using Red Hat Lightspeed .
- You can start conversion to a RHEL system using Red Hat Lightspeed .



NOTE

The only Red Hat Lightspeed service you can use with registered CentOS Linux systems is the RHEL conversion. All other Red Hat Lightspeed services are available only after the conversion to RHEL.

Additional resources

- How to configure HTTP proxy for Remote Host Configuration (rhc)
- How to configure HTTP Proxy for Red Hat Subscription Management
- How to apply a system wide proxy

4.4. ADDITIONAL RESOURCES

- RHEL Developer Subscription for Individuals

- RHEL subscription
- Red Hat Sales team

CHAPTER 5. REVIEW THE PRE-CONVERSION ANALYSIS REPORT USING RED HAT LIGHTSPEED

To assess if you can convert your CentOS Linux systems to RHEL, run the **Pre-conversion analysis for converting to RHEL** task. The pre-conversion analysis generates a report that summarizes potential problems and suggests recommended solutions.

The report also helps you decide whether it is possible or advisable to proceed with the conversion to RHEL.

Prerequisites

- You have completed the steps listed in Preparing for a RHEL conversion using Red Hat Lightspeed.

Procedure

1. Access the Red Hat Hybrid Cloud Console, and navigate to **Automation toolkit → Tasks**.
2. Locate the **Pre-conversion analysis for converting to RHEL**task and click **Select systems**.
3. Alternatively, log in to the Red Hat Hybrid Cloud Console , go to **Red Hat Enterprise Linux → Red Hat Lightspeed for RHEL → Inventory → Systems**, choose a system you want to convert and click **Convert system to RHEL** label.
4. In the **Task name** field type the name of the task and select the CentOS Linux 7 systems that you want to analyze for conversion. Click **Next**.
5. Configure the pre-conversion analysis task with the following settings:

   Do not use the ELS subscription

   Choose this option if you plan to upgrade your RHEL system to version 8 or higher.

   Allow kernel modules outside of RHEL repositories on the system

   Choose this option to allow the pre-conversion analysis to ignore kernel modules that are not part of RHEL repositories.

   Allow outdated kernel on the system

   Choose this option to allow the pre-conversion analysis to ignore when your system is booted from an outdated kernel.

   Allow outdated packages on the system

   Choose this option to allow the pre-conversion analysis to ignore all outdated packages on the system.

6. Click **Run task**. The pre-conversion analysis can take up to an hour to complete.
   The pre-conversion analysis utility generates a new report in the **Activity** tab. Select the report to view a summary of issues found in each system. You can also review further by selecting a system to view each issue and, when applicable, a potential remediation in detail.

   Each issue has a severity level:

   Inhibitor

   Would cause the conversion to fail because it is very likely to result in a deteriorated system state. You must resolve this issue before converting.

Overridable inhibitor

Would cause the conversion to fail because it is very likely to result in a deteriorated system state. You must resolve or manually override this issue before converting. For more details about inhibitors that you can override, see the step about configuring the settings of the pre-conversion analysis in this procedure.

Skipped

Could not run this test because of a prerequisite test failing. Could cause the conversion to fail.

Warning

Would not cause the conversion to fail. System and application issues might occur after the conversion.

Info

Informational with no expected impact to the system or applications.

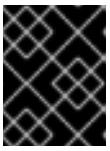
7. After reviewing the report and resolving all reported issues, click **Run task again** to rerun the analysis and confirm that there are no issues outstanding.

CHAPTER 6. CONVERT TO A RHEL SYSTEM BY USING RED HAT LIGHTSPEED

After running the **Pre-conversion analysis for converting to RHEL**task and resolving all reported issues, you can convert your CentOS Linux 7 systems to RHEL 7.

Prerequisites

- You have completed the steps listed in Preparing for a RHEL conversion using Red Hat Lightspeed and Reviewing the pre-conversion analysis report using Red Hat Lightspeed .

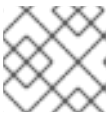


IMPORTANT

To avoid serious problems after the conversion, do not convert any systems that have unresolved inhibitors and warnings in the pre-conversion analysis.

Procedure

1. Access the Red Hat Hybrid Cloud Console, and navigate to **Automation toolkit → Tasks**.
2. Locate the **Convert to RHEL from CentOS 7 Linux**task and click **Run task**.
3. Select the CentOS Linux 7 systems that you want to convert to RHEL and click **Execute task**.



NOTE

The conversion process can take up to an hour to complete.

4. Go to the **Activity** tab and select the newly generated conversion report.
5. Review each system and message:
   - If a system has been successfully converted with no issues, reboot the system and proceed to the next step.
   - If the system was not converted, review the message for more information on the found problems and how to resolve them. Additionally, verify the following:
     - You have completed all steps in Preparing for a RHEL conversion using Red Hat Lightspeed.
     - The system has all packages required for the conversion.
     - The system is running.
     - You have resolved all issues found in the Reviewing the pre-conversion analysis report using Red Hat Lightspeed.
   - If the conversion timed out without completing, verify that the system is running and retry at another time. If issues persist, contact Support.
6. After rebooting the system that has been successfully converted, remove third-party packages from the original operating system that remained unchanged. These are typically packages that do not have a RHEL counterpart. To get a list of these packages, use:
   -

```
# yum list extras --disablerepo="" --enablerepo=<RHEL_RepoID>*
```

Replace *RHEL_RepoID* with your repository.

7. Optional: Perform an in-place upgrade to RHEL 9 to ensure your system has update with the latest enhancements, security features, and bug fixes. For more information, see the Upgrading from RHEL 7 to RHEL 8 and Upgrading from RHEL 8 to RHEL 9 guides.

CHAPTER 7. TROUBLESHOOTING CONVERSIONS

This chapter lists troubleshooting resources and tips.

7.1. TROUBLESHOOTING RESOURCES

To help you troubleshoot issues that can occur during the conversion process, review the log messages that you can find in the console and log files.

Console Output

By default, only info, warning, error, and critical log level messages are printed to the console output by the **Convert2RHEL** utility. To also print debug messages, use the **--debug** option with the **convert2rhel** command.

Logs

- The **/var/log/convert2rhel/convert2rhel.log** file lists debug, info, warning, error, and critical messages.
- The **/var/log/convert2rhel/rpm_va.log** file lists all package files on the unconverted system that a user has modified. The **rpm -Va** command generates this output. It is run automatically unless you use the **--no-rpm-va** option with the **convert2rhel** command.

7.2. FIX DEPENDENCY ERRORS

During a conversion from a different Linux distribution to RHEL, you can install certain packages without some of their dependencies.

Prerequisites

- You have successfully completed the conversion to RHEL. See Converting to a RHEL system for more information.

Procedure

1. Identify dependencies errors:

   ```
   # yum check dependencies
   ```

   If the command displays no output, no further actions are required.

2. To fix dependency errors, reinstall the affected packages. During this operation, the **yum** utility automatically installs missing dependencies. If the repositories that are available on the system do not have required dependencies, install those packages manually.

7.3. TROUBLESHOOTING ISSUES WITH RED HAT LIGHTSPEED CONVERSIONS

The following issues might occur when using Red Hat Lightspeed to convert to RHEL.

7.3.1. Fix missing systems in pre-conversion analysis task

When running the **Pre-conversion analysis for converting to RHEL**task in Red Hat Lightspeed, CentOS Linux 7 systems that appeared correctly in **RHEL Inventory** might not appear in the list of available systems to run the pre-conversion analysis on. This issue occurs when the Remote Host Configuration (RHC) is disconnected.

Procedure

1. Log in to the Red Hat Hybrid Cloud Console and go to **Red Hat Lightspeed → RHEL → Inventory → Systems**.
2. Select the affected system from the table.
3. In the **General Information** tab, go to the **System Status** card and verify the RHC status:
   a. If the RHC status is **Connected**, RHC is connected correctly.
   b. If the RHC status is **Not available**, RHC is disconnected. Proceed to the next step to reconnect RHC.
4. Unregister the system in your terminal:

```
# rhc disconnect
```

5. To help with troubleshooting, set the **RHC systemd** service (**rhcd**) logging to the highest level:

```
# sed -ie 's%error%trace%' /etc/rhc/config.toml
```

6. Register your system with Red Hat Lightspeed and re-enable RHC in your terminal:

```
# insights-client --register
# rhc connect -a <activation_key> -o <organization_ID>
```

   Replace *activation_key* and *organization_ID* with the activation key and organization ID from the Red Hat Customer Portal.

Verification

- Verify that you can select the system in the Pre-conversion analysis for converting to RHEL task.
  If the system still does not appear correctly, review error messages from **rhcd** and the **insights-client** tool:

```
# journalctl -u rhcd
# less /var/log/insights-client/insights-client.log
```

7.3.2. Pre-conversion analysis task fails to complete

After running the **Pre-conversion analysis for converting to RHEL**task, one or more of the systems can fail to generate a report with the error message **Task failed to complete for an unknown reason. Retry this task at a later time.** If this issue occurs, complete the following steps to troubleshoot.

Procedure

1. Verify if the affected system is unavailable, for example because of a network accessibility issue or because the system is shut off.
2. Review the **RHC systemd** service (**rhcd**) for errors:
   a. Stop rhcd in your terminal:

```
# systemctl stop rhcd
```

   b. Set **rhcd** logging to the highest level:

```
# sed -ie 's%error%trace%' /etc/rhc/config.toml
```

   c. Restart **rhcd**:

```
# systemctl start rhcd
```

   d. Review error messages posted by **rhcd**:

```
# journalctl -u rhcd
```

3. Review the **rhc-worker-script** log file for errors:

```
# less /var/log/rhc-worker-script/rhc-worker-script.log
```

7.4. KNOWN ISSUES AND LIMITATIONS

Learn about known issues and limitations that can occur during the conversion.

Conversion with Red Hat Content Delivery Network (CDN) or Satellite through the Red Hat Subscription Manager (RHSM)

Systems that connect to the Internet by using an HTTP proxy server cannot convert using Red Hat CDN or Satellite through RHSM by using the command line. To work around this problem, enable HTTP proxy for yum and then configure the HTTP proxy for RHSM:

1. Configure yum to use an HTTP proxy. For more information, see the Red Hat Knowledgebase solution How to enable proxy settings for yum command on RHEL?
2. Install the **subscription-manager** package:
   a. Download the Red Hat GPG key:

```
# curl -o /etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release
https://security.access.redhat.com/data/fd431d51.txt
```

   b. Install a repository file for the **client-tools** repository that has the **subscription-manager** package:
      - For conversions to RHEL 7:

```
# curl -o /etc/yum.repos.d/client-tools.repo https://cdn-
public.redhat.com/content/public/repofiles/client-tools-for-rhel-7-server.repo
```

- For conversions to RHEL 8:

```
# curl -o /etc/yum.repos.d/client-tools.repo https://cdn-public.redhat.com/content/public/repofiles/client-tools-for-rhel-8.repo
```

- For conversions to RHEL 9:

```
# curl -o /etc/yum.repos.d/client-tools.repo https://cdn-public.redhat.com/content/public/repofiles/client-tools-for-rhel-9.repo
```

c. If you are converting to an earlier version of RHEL 8, for example, RHEL 8.5, update the **$releasever** value in the client-tools repository:

```
# sed -i 's%\$releasever%<release_version>%' /etc/yum.repos.d/client-tools.repo
```

Replace *release_version* with the correct release version, for example **8.5** or **8.10**.

d. Install the following subscription-manager packages:

```
# yum -y install subscription-manager subscription-manager-rhsm-certificates
```

3. Configure HTTP proxy for RHSM. For more information, see the Red Hat Knowledgebase solution How to configure HTTP Proxy for Red Hat Subscription Management .

4. Register the system with RHSM:

```
# subscription-manager register --org <organization_id> --activationkey <activation_key>
```

Replace *organization_id* and *activation_key* with the organization ID and activation key from the Red Hat Customer Portal.

5. Remove the organization ID and activation key from the **/etc/convert2rhel.ini** file.

6. Perform the conversion to RHEL:

```
# convert2rhel
```

(RHELC-559)

Conversion of UEFI systems with Secure Boot

UEFI systems with Secure Boot enabled are not supported for conversion. To work around this issue, complete the following steps:

1. Disable Secure Boot before the conversion. For more information, see the Red Hat Knowledgebase solution convert2rhel fails on UEFI systems with Secure Boot enabled .

2. Perform the conversion to RHEL.

3. If converting from Oracle Linux 7 or Alma Linux 8, install the **shim-x64** package:

```
# yum install -y shim-x64
```

4. Re-enable Secure Boot.

(RHELC-138, RHELC-1190)

Conversion with Red Hat Lightspeed

If you are converting by using Red Hat Lightspeed, running two RHC daemon (rhcd) processes simultaneously prevents the pre-conversion analysis from running as expected. To prevent this issue, run only one rhcd process at a time. (HMS-2629)

7.5. OBTAINING SUPPORT

If you experience problems during the conversion, notify Red Hat so that these problems can be addressed.



IMPORTANT

If you are experiencing problems during the conversion, raise a Support case of Severity 3 or Severity 4 level only. For more details, see Production Support Terms of Service.

Prerequisites

- The **sos** package is installed. You must use this package to generate an **sosreport** that the Red Hat Support team requires when you open a support case.

Procedure

- To obtain support, perform either of the following steps:
  - Open a support case:
    - Select the appropriate version of RHEL as the product, and attach an **sosreport** from your system.
    - Generate an **sosreport** on your system:

      ```
      # sosreport
      ```

      Note that you can leave the case ID empty.
  - Submit a bug report:
    - Open a bug, select the appropriate version of RHEL as the product, and select **convert2rhel** as the component.

Additional resources

- What is an sosreport and how to create one in Red Hat Enterprise Linux?
- How do I open and manage a support case on the Customer Portal?
- Convert2RHEL Support Policy

CHAPTER 8. ADDITIONAL RESOURCES

- Converting from a Linux distribution to RHEL using the Convert2RHEL utility
- How to perform an unsupported conversion from a RHEL-derived Linux distribution to RHEL
- Red Hat Enterprise Linux technology capabilities and limits
- Convert2RHEL FAQ (Frequently Asked Questions)